Convergence Rates of First-Order Methods: Proofs and Numerical Experiments

Arthur Conche
M2 MMAS, Université Paris Cité

Working draft — October 13, 2026

Abstract

This report studies the convergence of first-order optimization methods (gradient descent, Nesterov's accelerated gradient and stochastic gradient descent) for smooth functions. For each method we prove the classical convergence rate under convexity, strong convexity or the Polyak–Łojasiewicz condition, and we compare these theoretical bounds with numerical experiments. *[Draft: to be completed.]*

1 Introduction

[To be written: motivation, question studied, contributions, outline.]

2 Setting and assumptions

We consider the unconstrained problem

$$\min_{x \in \mathbb{R}^d} f(x),$$

where $f : \mathbb{R}^d \to \mathbb{R}$ is differentiable, and we denote by $f^* = \inf f$ and by x^* a minimizer when one exists.

Definition 1 (L-smoothness)**.** Let $L > 0$. A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is *L-smooth* if its gradient is L-Lipschitz:

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^d.$$

[To add: convexity, μ-strong convexity, condition number $\kappa = L/\mu$, Polyak–Łojasiewicz condition.]

3 Preliminary results

Lemma 1 (Descent lemma)**.** *Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable and L-smooth. Then for all $x, y \in \mathbb{R}^d$,*

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2 .$$

Proof. Since ∇f is L-Lipschitz, it is continuous, hence f is of class C^1. Fix $x, y \in \mathbb{R}^d$ and set $\varphi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. The function φ is C^1 with $\varphi'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle$, so by the fundamental theorem of calculus,

$$f(y) - f(x) = \varphi(1) - \varphi(0) = \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle \, dt.$$

Since $\int_0^1 \langle \nabla f(x), y - x \rangle \, dt = \langle \nabla f(x), y - x \rangle$, we obtain

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle = \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \, dt.$$

By the Cauchy–Schwarz inequality, then L-smoothness,

$$\begin{aligned}
\int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \, dt &\leq \int_0^1 \|\nabla f(x + t(y - x)) - \nabla f(x)\| \, \|y - x\| \, dt \\
&\leq \int_0^1 L \, \|t(y - x)\| \, \|y - x\| \, dt \\
&= L \, \|y - x\|^2 \int_0^1 t \, dt \qquad \text{(since } t \geq 0\text{)} \\
&= \frac{L}{2} \, \|y - x\|^2 ,
\end{aligned}$$

which gives the result. □

Remark 1. Convexity is not used in the proof: the descent lemma holds for every L-smooth function. This is what allows the analysis of gradient descent for non-convex functions satisfying the Polyak–Łojasiewicz condition (Section 4.3).

Remark 2. Since the Cauchy–Schwarz inequality bounds the absolute value of the integrand, the same argument gives the two-sided bound

$$|f(y) - f(x) - \langle \nabla f(x), y - x \rangle| \leq \frac{L}{2} \, \|y - x\|^2 \qquad \text{for all } x, y \in \mathbb{R}^d.$$

Corollary 1 (Sufficient decrease)**.** [To prove.] *If f is L-smooth and $x^+ = x - \frac{1}{L} \nabla f(x)$, then*

$$f(x^+) \leq f(x) - \frac{1}{2L} \, \|\nabla f(x)\|^2 .$$

4 Gradient descent

4.1 Convex case

[R1: $f(x_k) - f^* \leq \dfrac{L \, \|x_0 - x^*\|^2}{2k}$.]

4.2 Strongly convex case

[R2: $\|x_k - x^*\|^2 \leq (1 - 1/\kappa)^k \, \|x_0 - x^*\|^2$.]

4.3 Polyak–Łojasiewicz condition

[R4: linear rate without convexity.]

5 Nesterov's accelerated gradient

[R3: $O(1/k^2)$ via a Lyapunov function; $(1 - 1/\sqrt{\kappa})^k$ in the strongly convex case; lower bound; ODE interpretation.]

6 Stochastic gradient descent

[R5: constant step size and noise floor; decreasing step size and $O(1/k)$ rate.]

7 Numerical experiments

[Quadratics with controlled κ; logistic regression; non-convex PL example; SGD on least squares; comparison with Adam.]

8 Discussion

[Where the bounds are tight, where they are pessimistic, and why.]

References

[1] G. Garrigos and R. M. Gower. *Handbook of Convergence Theorems for (Stochastic) Gradient Methods*. arXiv:2301.11235, 2023.

[2] S. Bubeck. *Convex Optimization: Algorithms and Complexity*. Foundations and Trends in Machine Learning, 2015.

[3] Y. Nesterov. *Lectures on Convex Optimization*. Springer, 2nd edition, 2018.

[4] H. Karimi, J. Nutini and M. Schmidt. Linear convergence of gradient and proximal-gradient methods under the Polyak–Łojasiewicz condition. *ECML PKDD*, 2016.

[5] W. Su, S. Boyd and E. J. Candès. A differential equation for modeling Nesterov's accelerated gradient method. *NeurIPS*, 2014.